Priya Kumari

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Education

Indian Institute of Technology (IIT) Mandi Sep 2024 – May 2026

- **M.Tech** – Computer Science & Engineering (Pursuing)

YMCA University of Science & Technology, Faridabad Sep 2018 – May2022

- **B.Tech** – Computer Science & Engineering | **CGPA: 7.73**

Skills

- **Languages**: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS, C
- **Frameworks**: React, Node.js, Express.js, Spring Boot, Vite, Redux Toolkit
- **AI / ML**: PyTorch, TensorFlow, OpenCV, Scikit-learn, LLM APIs, LangChain, Multi-Agent Systems
- **Databases**: MongoDB, MySQL
- **Cloud & Tools**: Google Cloud Platform (GCP), Docker, Git, VS Code, IntelliJ
- **Relevant subject**: DeepLearning,MachingLearning, DBMS, OOPS, DataStructure And Algorithm

Work Experience

Bosch Global Software Technologies AI / ML Intern Bengaluru, India Jan 2026 – Present

- **Multi-Agent AI Systems:**
  - Built Agent-to-Agent Protocol–based AI systems for enterprise automation; designed scalable multi-agent LLM communication workflows.
  - Integrated LLM APIs into production-grade ML pipelines; contributed to distributed backend services and system design.

QuadBtech Technology Front End Developer May 2022 – Aug202

- **Frontend Development:**
  - Built responsive e-commerce platform with React & Bootstrap; implemented product listings, cart, and authentication features.
  - Developed daily report visualizations and optimized UI performance for improved cross-device user experience.

IT Shaala Java Backend TraineeRemote Mar 2023 – Aug 2023

- **Backend Development:**
  - Built scalable backend apps with Spring Boot, Hibernate, REST APIs; managed MySQL and NoSQL databases in team projects.

Projects

RAG – AUTOSAR Document Intelligence System | Python · Qdrant · GLM · VLM · BM25 · Dense Retrieval · Hybrid Retrieval · Ollama

- Built a production-ready RAG pipeline on open-source AUTOSAR documentation with end-to-end document understanding using layout-aware OCR via GLM for section detection across complex technical PDFs.
- Designed a type-aware chunking strategy keeping text, table, and image chunks strictly separate — leveraging a VLM for image understanding and an LLM for semantic table extraction to preserve data integrity.
- Deployed hybrid retrieval combining dense vector search and BM25 via Qdrant, significantly improving retrieval precision over single-method approaches.
- Powered the generation layer with open-source **Ollama LLMs** — fully local, zero external API dependency, ensuring data privacy and offline operability.

Video Conferencing App (Link) **:-*React.js, Node.js, WebRTC, Socket.IO, Express.js***

- Real-time distributed system with group video/audio, integrated chat, secure auth, and scalable REST & WebSocket APIs via Socket.IO.

Tuberculosis Detection – Deep Learning (Link) :- ***PyTorch, OpenCV, Transformer, WSI***

- Automated ML pipeline for TB detection on Whole Slide Images using PyTorch, Transformer-based weakly supervised model, and heatmap interpretability.

Stock Trading Platform – Zerodha Clone (Link) **:-*React, Node.js, Express.js, MongoDB, TypeScript***

- Full-stack multi-tier architecture with React frontend, Express.js + MongoDB backend, secure auth, live dashboard, and REST API service boundaries.

Certifications & Achievements

- **GATE 2023 & 2024 Qualified** — top national engineering postgraduate entrance exam.
- **Full Stack Web Dev (MERN)** | **React.js with TypeScript & State Management** | **Data Structures & Algorithms**
- **Java (NPTEL)** | **Python** | **Java Backend Development**